EXERCISE

Probability Mass Function (PMF)

Which values belong to the event?

x takes only the errors $-1, 0, 2$ cm, with masses $1/2, 1/4, 1/4$, respectively. What is $P(x \geq 0)$?

A $1/4$: count only positive errors.

B $1/2$: include zero and positive errors.

C $2/3$: two of three possible values.

Event probability; PMF — [1] DL §3.3.1 · Course question

SOLUTION

Probability Mass Function (PMF)

Which values belong to the event?

Solution 1 / 3

Identify the event

x: error (cm); $p(a) = P(x = a)$: probability mass at value a.

---

1. Possible values: $-1, 0, 2$; masses: $1/2, 1/4, 1/4$.
2. Nonnegative means $x \geq 0$, so include 0 and 2.

Event probability; PMF — [1] DL §3.3.1 · Course question

SOLUTION

Probability Mass Function (PMF)

Which values belong to the event?

Solution 2 / 3

Add the included masses

x: error (cm); $p(a) = P(x = a)$: probability mass at value a.

---

1. $P(x \geq 0) = P(x = 0) + P(x = 2)$.
2. $= 1/4 + 1/4 = 2/4 = 1/2$.
3. Complement: $x < 0$ contains all outcomes outside $x \geq 0$; $P(x < 0) + P(x \geq 0) = 1$.

Event probability; PMF — [1] DL §3.3.1 · Course question

SOLUTION

Probability Mass Function (PMF)

Which values belong to the event?

Solution 3 / 3

Choose B

x: error (cm); $p(a) = P(x = a)$: probability mass at value a.

---

1. A excludes zero and answers a different event.
2. B includes both nonnegative values and their masses.
3. C treats unequal masses as equal. Count probability, not values.

Event probability; PMF — [1] DL §3.3.1 · Course question

RESULT

Probability Mass Function (PMF)

Which values belong to the event?

x takes only the errors $-1, 0, 2$ cm, with masses $1/2, 1/4, 1/4$, respectively. What is $P(x \geq 0)$?

A $1/4$: count only positive errors.

B $1/2$: include zero and positive errors. ✓

C $2/3$: two of three possible values.

> $P(x \geq 0) = p(0) + p(2) = 1/4 + 1/4 = 1/2$. Total mass is 1.

Event probability; PMF — [1] DL §3.3.1 · Course question

EXERCISE

Probability Density Function (PDF)

Density is height; probability is area

x is a scalar error, uniform on $[0, .5]$ cm with zero density outside. Which statement is valid?

A. Density 2 cm^{-1}; $P(.1 \leq x \leq .3) = .4$.

B. Density 2 means event probability 2.

C. Probability is $.2$, the interval width alone.

Density; event probability — [1] DL §3.3.2 · Course question

SOLUTION

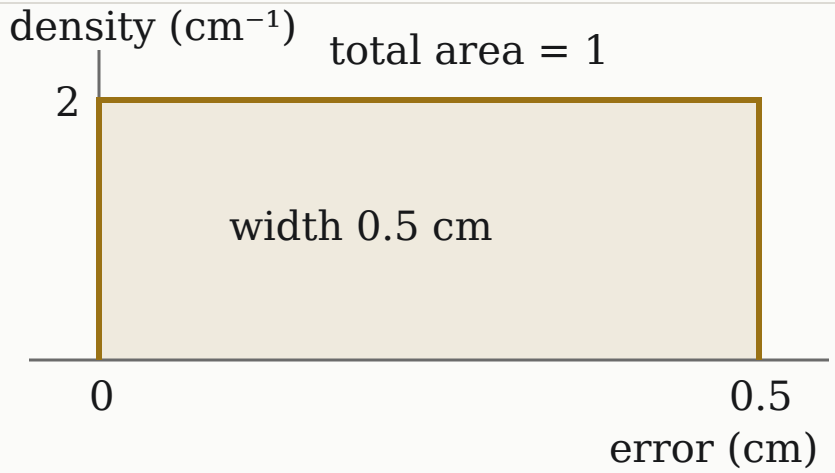
Probability Density Function (PDF)

Density is height; probability is area

Normalize the density

x: error (cm); h: uniform density; t: integration variable (cm).

1. Uniform support: $[0, .5]$ cm; width $.5$ cm.
2. Normalization means total area is one. $h(.5 - 0) = 1$.
3. $h = 1/.5 = 2\ \text{cm}^{-1}$.



Density; event probability — [1] DL §3.3.2 · Course question

SOLUTION

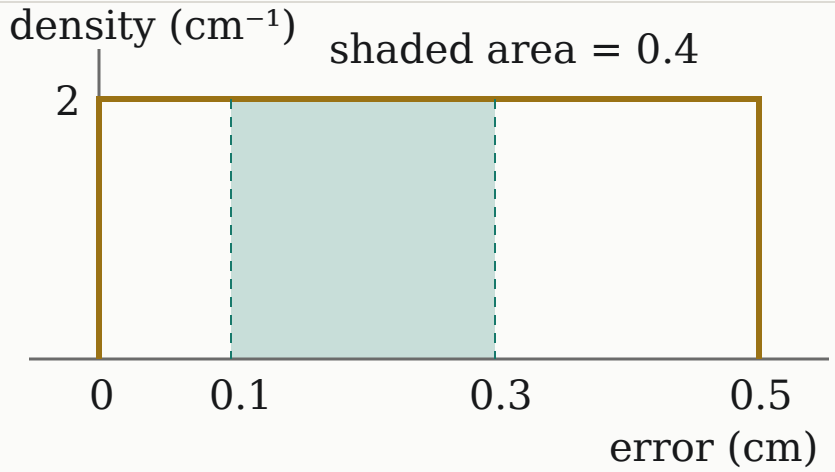
Probability Density Function (PDF)

Density is height; probability is area

Integrate over the requested interval

x: error (cm); h: uniform density; t: integration variable (cm).

1. $P(.1 \le x \le .3) = \int_{.1}^{.3} h\,dt.$
2. $= 2(.3 - .1) = 2(.2) = .4.$
3. Density times width has units $\text{cm}^{-1}\text{cm} = 1.$



Density; event probability — [1] DL §3.3.2 · Course question

SOLUTION

Probability Density Function (PDF)

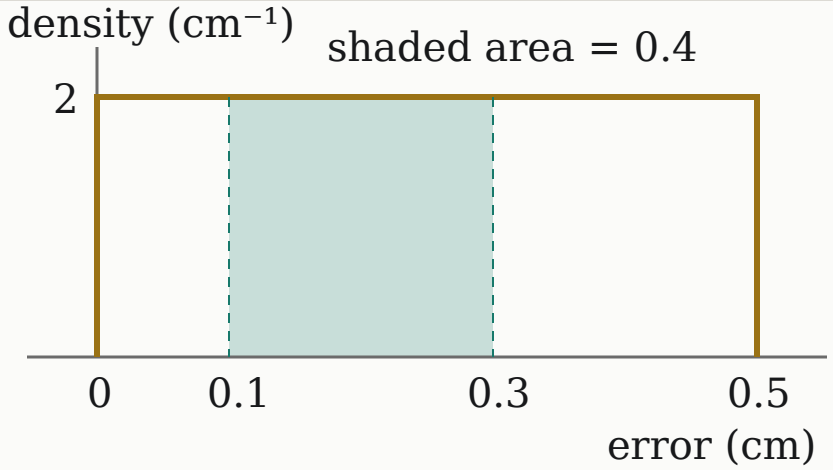
Density is height; probability is area

Solution 3 / 3

Choose A

x: error (cm); h: uniform density; t: integration variable (cm).

1. A gives both the normalized density and event probability.
2. B confuses density height with probability; probabilities cannot exceed 1.
3. C omits the height: width alone is not the area.



Density; event probability — [1] DL §3.3.2 · Course question

RESULT

Probability Density Function (PDF)

Density is height; probability is area

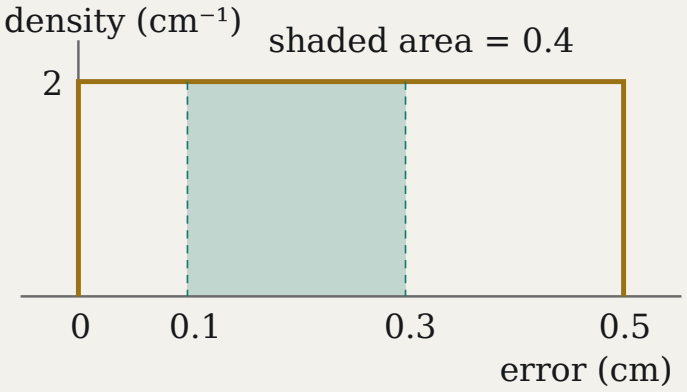
x is a scalar error, uniform on $[0, .5]$ cm with zero density outside. Which statement is valid?

A. Density 2 cm^{-1}; $P(.1 \le x \le .3) = .4$. ✓

B. Density 2 means event probability 2.

C. Probability is .2, the interval width alone.

$h(.5) = 1 \Rightarrow h = 2$; mass $= h(.3 - .1) = .4$.



Density; event probability — [1] DL §3.3.2 · Course question

EXERCISE

Moments

Variance is the average squared deviation from the mean.

For scalar error x (cm), $\mu = \mathbb{E}[x] = 1/4$ cm and $\mathbb{E}[x^2] = 5/4$ cm^2. Using $\mathrm{Var}(x) = \mathbb{E}[(x-\mu)^2]$, choose the variance.

A. $5/4$: the second moment is always variance.

B. 1: subtract the mean, not its square.

C. $19/16$: subtract the squared mean.

Centering; variance — [1] DL §3.8 · Course question

SOLUTION

Moments

Variance is the average squared deviation from the mean.

Solution 1 / 3

Expand the centered square

x: error; $\mathbb{E}$: expectation; $\mu = \mathbb{E}[x]$: mean; Var: variance.

1. x: error; $\mu = \mathbb{E}[x] = 1/4$ cm.
2. $\mathbb{E}[x^2] = 5/4$ cm^2 is the second moment.
3. $(x - \mu)^2 = x^2 - 2\mu x + \mu^2$.

Centering; variance — [1] DL §3.8 · Course question

SOLUTION

Moments

Variance is the average squared deviation from the mean.

Solution 2 / 3

Take expectation, then substitute

x: error; $\mathbb{E}$: expectation; $\mu = \mathbb{E}[x]$: mean; Var: variance.

1. $\mathrm{Var}(x) = \mathbb{E}[x^2] - 2\mu\mathbb{E}[x] + \mu^2.$
2. $\mathbb{E}[x] = \mu \;\Rightarrow\; \mathrm{Var}(x) = \mathbb{E}[x^2] - \mu^2.$
3. $= 5/4 - (1/4)^2 = 20/16 - 1/16 = 19/16 \text{ cm}^2.$

Centering; variance — [1] DL §3.8 · Course question

SOLUTION

Moments

Variance is the average squared deviation from the mean.

Solution 3 / 3

Choose C

x: error; $\mathbb{E}$: expectation; $\mu = \mathbb{E}[x]$: mean; Var: variance.

---

1. A leaves the squared mean inside the second moment.
2. B subtracts a length from a squared length; its units do not match.
3. C subtracts the squared mean. Standard deviation $\sqrt{\mathrm{Var}(x)}$ measures spread in cm.

Centering; variance — [1] DL §3.8 · Course question

RESULT

Moments

Variance is the average squared deviation from the mean.

For scalar error x (cm), $\mu = \mathbb{E}[x] = 1/4$ cm and $\mathbb{E}[x^2] = 5/4$ cm^2. Using $\mathrm{Var}(x) = \mathbb{E}[(x-\mu)^2]$, choose the variance.

A $5/4$: the second moment is always variance.

B 1: subtract the mean, not its square.

C $19/16$: subtract the squared mean. ✓

> $\mathrm{Var}(x) = \mathbb{E}[x^2] - \mu^2 = 5/4 - 1/16 = 19/16$ cm^2.

Centering; variance — [1] DL §3.8 · Course question

EXERCISE

Moments

Independent sources, dependent components

U, V are independent, dimensionless signs: $P(U = u) = P(V = u) = 1/2$ for $u = \pm 1$. Given $\mathbb{E}[U] = \mathbb{E}[V] = 0$, $\mathrm{Var}(U) = \mathrm{Var}(V) = 1$, $\mathrm{Cov}(U, V) = 0$. For $x = (U, 2U + V)^T$, find $\mathrm{Cov}(x_1, x_2)$.

A. 2: both components share U.

B. 0: independent sources imply independent outputs.

C. 1: doubling a coefficient has no effect.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · [1] DL §2.7 · Course question

SOLUTION

Moments

Independent sources, dependent components

Separate sources from components

U, V: independent fair signs; $x = (U, 2U + V)^T$; $\Sigma_{ij} = \mathrm{Cov}(x_i, x_j)$.

1. U, V: independent signs, each equally likely -1 or 1.
2. $\mathbb{E}[U] = \mathbb{E}[V] = (-1 + 1)/2 = 0$; hence $\mathbb{E}[x_1] = \mathbb{E}[x_2] = 0$.
3. $\mathbb{E}[U^2] = \mathbb{E}[V^2] = 1$; $\mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] = 0$.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · [1] DL §2.7 · Course question

SOLUTION

Moments

Independent sources, dependent components

Compute cross-covariance

U, V: independent fair signs; $x = (U, 2U + V)^T$; $\Sigma_{ij} = \mathrm{Cov}(x_i, x_j)$.

1. $\Sigma_{12} = \mathrm{Cov}(x_1, x_2) = \mathbb{E}[x_1 x_2]$.
2. $= \mathbb{E}[U(2U + V)] = 2\mathbb{E}[U^2] + \mathbb{E}[UV]$.
3. $= 2(1) + 0 = 2$.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · [1] DL §2.7 · Course question

SOLUTION

Moments

Independent sources, dependent components

Check the complete covariance

U, V: independent fair signs; $x = (U, 2U + V)^T$; $\Sigma_{ij} = \mathrm{Cov}(x_i, x_j)$.

1. $\Sigma_{11} = \mathbb{E}[U^2] = 1.$
2. $\Sigma_{22} = \mathbb{E}[(2U + V)^2] = 4(1) + 4(0) + 1 = 5.$
3. $\Sigma = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$; $\det \Sigma = 1(5) - 2(2) = 1.$
4. For a direction column $a \in \mathbb{R}^2$, positive definite means $a^T \Sigma a > 0$ whenever $a \neq 0$.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · [1] DL §2.7 · Course question

SOLUTION

Moments

Independent sources, dependent components

Solution 4 / 4

Choose A

U, V: independent fair signs; $x = (U, 2U + V)^T$; $\Sigma_{ij} = \mathrm{Cov}(x_i, x_j)$.

1. A retains the shared source U.
2. B confuses independence of U and V with independence of their outputs.
3. C ignores the coefficient 2 in the shared contribution.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · [1] DL §2.7 · Course question

RESULT

Moments

Independent sources, dependent components

U, V are independent, dimensionless signs: $P(U = u) = P(V = u) = 1/2$ for $u = \pm 1$. Given $\mathbb{E}[U] = \mathbb{E}[V] = 0$, $\mathrm{Var}(U) = \mathrm{Var}(V) = 1$, $\mathrm{Cov}(U, V) = 0$. For $x = (U, 2U + V)^T$, find $\mathrm{Cov}(x_1, x_2)$.

A 2: both components share U. ✓

B 0: independent sources imply independent outputs.

C 1: doubling a coefficient has no effect.

$\mathbb{E}[U(2U + V)] = 2$; $\Sigma = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$, $\det \Sigma = 1$.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · [1] DL §2.7 · Course question

EXERCISE

Gaussian geometry

Which point has the higher Gaussian density?

$\mu = 0$, $\Sigma = \mathrm{diag}(4, 1)$; $u = (2, 0)^T$, $v = (0, 2)^T$. Compare Gaussian densities.

u, v: fixed 2×1 points; μ: fixed mean column; Σ: fixed 2×2 covariance; p: density (inverse squared coordinate units). Coordinates use original units; covariance entries use squared units.

A $p(u) = p(v)$: their Euclidean lengths agree.

B $p(u) > p(v)$: u is one standard deviation away.

C $p(u) < p(v)$: higher variance always lowers density.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

SOLUTION

Gaussian geometry

Which point has the higher Gaussian density?

Solution 1 / 3

Use the covariance-scaled distance

u, v: fixed point columns; $x = (x_1, x_2)^T$: generic point. μ: mean; Σ: covariance; p: density (inverse squared coordinate units); d_M: dimensionless covariance-scaled distance. Coordinate units / squared covariance units.

---

1. $\mu = 0$: mean; $\Sigma = \mathrm{diag}(4, 1)$: covariance.
2. $u = (2, 0)^T$, $v = (0, 2)^T$ are points in the same Gaussian.
3. $d_M^2(x) = (x - \mu)^T \Sigma^{-1} (x - \mu) = x_1^2/4 + x_2^2$.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

SOLUTION

Gaussian geometry

Which point has the higher Gaussian density?

Compare exponents

u, v: fixed point columns; $x = (x_1, x_2)^T$: generic point. μ: mean; Σ: covariance; p: density (inverse squared coordinate units); d_M: dimensionless covariance-scaled distance. c: fixed normalization constant, same units as p; e: natural-exponential base. Coordinate units / squared covariance units.

---

1. $d_M^2(u) = 2^2/4 + 0^2 = 1$; $d_M^2(v) = 0^2/4 + 2^2 = 4$.
2. $p(x) = c\exp(-d_M^2(x)/2)$; the normalization c is shared.
3. $p(u)/p(v) = e^{-1/2}/e^{-4/2} = e^{3/2} > 1$.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

SOLUTION

Gaussian geometry

Which point has the higher Gaussian density?

Solution 3 / 3

Choose B

u, v: fixed point columns; $x = (x_1, x_2)^T$: generic point. μ: mean; Σ: covariance; p: density (inverse squared coordinate units); d_M: dimensionless covariance-scaled distance. Coordinate units / squared covariance units.

---

1. A uses ordinary distance and ignores different directional variances.
2. B: u is one standard deviation away; v is two.
3. C confuses changing covariance with comparing points in one fixed model.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

RESULT

Gaussian geometry

Which point has the higher Gaussian density?

$\mu = 0$, $\Sigma = \mathrm{diag}(4, 1)$; $u = (2, 0)^T$, $v = (0, 2)^T$. Compare Gaussian densities.

u, v: fixed 2×1 points; μ: fixed mean column; Σ: fixed 2×2 covariance; p: density (inverse squared coordinate units). Coordinates use original units; covariance entries use squared units.

A $p(u) = p(v)$: their Euclidean lengths agree.

B $p(u) > p(v)$: u is one standard deviation away. ✓

C $p(u) < p(v)$: higher variance always lowers density.

$d_M^2(u) = 1$, $d_M^2(v) = 4$; $p(u)/p(v) = e^{3/2} > 1$.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

EXERCISE

Bayes

How does prevalence change the chance an alarm is true?

$P(H) = .1$, $P(+ \mid H) = .9$, $P(+ \mid \neg H) = .1$. What is $P(H \mid +)$?

H: event; $\neg H$: no event; $+$: positive reading; P: probability (no units); $\mid$: given.

A. .9: the chance of a true event equals sensor sensitivity.

B. .1: an observation cannot change the prior.

C. .5: true and false positives contribute equally.

Prior probability; posterior probability — [1] DL §3.11 · Course question

SOLUTION

Bayes

How does prevalence change the chance an alarm is true?

Name the conditional populations

H: event; $\neg H$: no event; $+$: positive reading; P: probability (no units); $|$: given; $\cap$: both occur.

1. $P(H) = .1$, $P(\neg H) = 1 - .1 = .9$.
2. $P(+ \mid H) = .9$; $P(+ \mid \neg H) = .1$.

Prior probability; posterior probability — [1] DL §3.11 · Course question

SOLUTION

Bayes

How does prevalence change the chance an alarm is true?

Solution 2 / 3

Find all positives, then normalize

H: event; $\neg H$: no event; $+$: positive reading; P: probability (no units); $|$: given; $\cap$: both occur.

1. $P(H \cap +) = P(+ \mid H)P(H) = .9(.1) = .09.$
2. $P(\neg H \cap +) = P(+ \mid \neg H)P(\neg H) = .1(.9) = .09.$
3. $P(+) = .09 + .09 = .18$; $P(H \mid +) = .09/.18 = .5.$

Prior probability; posterior probability — [1] DL §3.11 · Course question

SOLUTION

Bayes

How does prevalence change the chance an alarm is true?

Solution 3 / 3

Choose C

H: event; $\neg H$: no event; $+$: positive reading; P: probability (no units); $|$: given; $\cap$: both occur.

1. A reverses the conditioning: sensitivity starts with event cases.
2. B uses the prior and ignores the positive observation.
3. C: true and false positives have equal probability mass.

Prior probability; posterior probability — [1] DL §3.11 · Course question

RESULT

Bayes

How does prevalence change the chance an alarm is true?

$P(H) = .1$, $P(+ \mid H) = .9$, $P(+ \mid \neg H) = .1$. What is $P(H \mid +)$?

H: event; $\neg H$: no event; $+$: positive reading; P: probability (no units); $\mid$: given.

A $.9$: the chance of a true event equals sensor sensitivity.

B $.1$: an observation cannot change the prior.

C $.5$: true and false positives contribute equally. ✓

$$P(H \mid +) = \frac{.9(.1)}{.9(.1) + .1(.9)} = .5.$$

Prior probability; posterior probability — [1] DL §3.11 · Course question

EXERCISE

Gaussian conditioning

What does a noisier reading change?

Scalar $x \sim \mathcal{N}(0, 4)$; observe $z = x + \varepsilon = 3$, with $\varepsilon \sim \mathcal{N}(0, R)$ independent of x. Known noise variance $R : 1 \to 4$. Compare posterior mean and variance.

x: scalar state; z: fixed reading; R: independent noise variance. $\mathcal{N}$: normal distribution, mean then variance; states/readings use state units and variances use squared units.

A. Mean moves toward 0; posterior variance increases.

B. Mean moves toward 3; posterior variance decreases.

C. Only the mean changes; variance ignores sensor noise.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

SOLUTION

Gaussian conditioning

What does a noisier reading change?

State the scalar measurement model

x, z, ε: scalar state, fixed reading and independent noise (state units). μ_0, P_0: prior mean/variance; μ_1, P_1: posterior mean/variance; R: noise variance (squared units); K: dimensionless gain.

---

1. $x \sim \mathcal{N}(\mu_0, P_0)$ with $\mu_0 = 0$, $P_0 = 4$.
2. $z = x + \varepsilon = 3$; independent noise $\varepsilon \sim \mathcal{N}(0, R)$.
3. $K = P_0/(P_0 + R)$; $\mu_1 = \mu_0 + K(z - \mu_0)$; $P_1 = (1 - K)P_0$.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

SOLUTION

Gaussian conditioning

What does a noisier reading change?

Compare the two noise variances

x, z, ε: scalar state, fixed reading and independent noise (state units). μ_0, P_0: prior mean/variance; μ_1, P_1: posterior mean/variance; R: noise variance (squared units); K: dimensionless gain.

1. Old $R = 1$: $K = 4/5$, $\mu_1 = 0 + (4/5)3 = 2.4$, $P_1 = (1/5)4 = .8$.
2. New $R = 4$: $K = 4/8 = 1/2$, $\mu_1 = 0 + (1/2)3 = 1.5$.
3. New $P_1 = (1 - 1/2)4 = 2$; precision check: $P_1^{-1} = 1/4 + 1/4 = 1/2$.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

SOLUTION

Gaussian conditioning

What does a noisier reading change?

Solution 3 / 3

Choose A

x, z, ε: scalar state, fixed reading and independent noise (state units). μ_0, P_0: prior mean/variance; μ_1, P_1: posterior mean/variance; R: noise variance (squared units); K: dimensionless gain.

1. A: the mean is closer to 0 and posterior variance rises from .8 to 2.
2. B reverses both changes: a noisier reading carries less information.
3. C ignores R in the posterior variance. The sensor value z did not change.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

RESULT

Gaussian conditioning

What does a noisier reading change?

Scalar $x \sim \mathcal{N}(0, 4)$; observe $z = x + \varepsilon = 3$, with $\varepsilon \sim \mathcal{N}(0, R)$ independent of x. Known noise variance $R : 1 \to 4$. Compare posterior mean and variance.

x: scalar state; z: fixed reading; R: independent noise variance. $\mathcal{N}$: normal distribution, mean then variance; states/readings use state units and variances use squared units.

- **A** Mean moves toward 0; posterior variance increases. ✓
- **B** Mean moves toward 3; posterior variance decreases.
- **C** Only the mean changes; variance ignores sensor noise.

> $K = 4/(4+4) = 1/2$; $\mu_1 = 1.5$, $P_1 = 2$ (before: $2.4, .8$).

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

EXERCISE

Gaussian conditioning

Does the conditional variance change?

Scalars a, b are jointly Gaussian with mean $\mu = (1, 2)^T$ and covariance $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Keep this model fixed. Observe $b = -1$ instead of 5 and compare the conditional mean and variance of $a \mid b$.

a: unknown scalar; b: observed scalar; μ: fixed 2×1 joint mean; Σ: fixed 2×2 covariance. Values use coordinate units; covariances use paired unit products.

A. Both conditional mean and variance decrease.

B. Mean decreases; conditional variance stays fixed.

C. Neither changes because the joint model is fixed.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

SOLUTION

Gaussian conditioning

Does the conditional variance change?

Read the Gaussian blocks

a, b: unknown/observed scalars; μ_a, μ_b: marginal means; Σ_{aa}, Σ_{bb}: variances; $\Sigma_{ab} = \Sigma_{ba}$: cross-covariance; $\mu_{a|b}, \Sigma_{a|b}$: conditional mean/variance. Means use coordinate units; covariance entries use paired product units.

---

1. a: unobserved scalar; b: observed scalar.
2. $\mu_a = 1$, $\mu_b = 2$; $\Sigma_{aa} = 4$, $\Sigma_{ab} = \Sigma_{ba} = 2$, $\Sigma_{bb} = 9$.
3. $\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(b - \mu_b)$.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

SOLUTION

Gaussian conditioning

Does the conditional variance change?

Update the center

a, b: unknown/observed scalars; μ_a, μ_b: marginal means; Σ_{aa}, Σ_{bb}: variances; $\Sigma_{ab} = \Sigma_{ba}$: cross-covariance; $\mu_{a|b}, \Sigma_{a|b}$: conditional mean/variance. Means use coordinate units; covariance entries use paired product units.

---

1. Old $b = 5$: $\mu_{a|b} = 1 + (2/9)(5 - 2) = 1 + 2/3 = 5/3$.
2. New $b = -1$: $\mu_{a|b} = 1 + (2/9)(-1 - 2) = 1 - 2/3 = 1/3$.
3. The innovation $b - \mu_b$ changed; the mean decreases.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

SOLUTION

Gaussian conditioning

Does the conditional variance change?

Solution 3 / 4

Compute the unchanged variance

a, b: unknown/observed scalars; μ_a, μ_b: marginal means; Σ_{aa}, Σ_{bb}: variances; $\Sigma_{ab} = \Sigma_{ba}$: cross-covariance; $\mu_{a|b}, \Sigma_{a|b}$: conditional mean/variance. Means use coordinate units; covariance entries use paired product units.

---

1. $\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}.$
2. $= 4 - 2(1/9)2 = 36/9 - 4/9 = 32/9.$
3. This expression contains covariance blocks, not the observed value b.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

SOLUTION

Gaussian conditioning

Does the conditional variance change?

Solution 4 / 4

Choose B

a, b: unknown/observed scalars; μ_a, μ_b: marginal means; Σ_{aa}, Σ_{bb}: variances; $\Sigma_{ab} = \Sigma_{ba}$: cross-covariance; $\mu_{a|b}, \Sigma_{a|b}$: conditional mean/variance. Means use coordinate units; covariance entries use paired product units.

---

1. A wrongly changes the width when only the observed value changes.
2. B changes the mean and keeps the conditional variance fixed.
3. C ignores the changed innovation. Constant width here uses joint Gaussianity.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

RESULT

Gaussian conditioning

Does the conditional variance change?

Scalars a, b are jointly Gaussian with mean $\mu = (1, 2)^T$ and covariance $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Keep this model fixed. Observe $b = -1$ instead of 5 and compare the conditional mean and variance of $a \mid b$.

a: unknown scalar; b: observed scalar; μ: fixed 2×1 joint mean; Σ: fixed 2×2 covariance. Values use coordinate units; covariances use paired unit products.

A Both conditional mean and variance decrease.

B Mean decreases; conditional variance stays fixed. ✓

C Neither changes because the joint model is fixed.

$\mu_{a|b} = 1 + \frac{2}{9}(-1 - 2) = \frac{1}{3}$; $\Sigma_{a|b} = 4 - \frac{4}{9} = \frac{32}{9}$.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

EXERCISE

Estimation

What makes the MAP estimate unique?

$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ known; $b = (1,2)^T$ observed; $\theta \in \mathbb{R}^2$. Model: $b \mid \theta \sim \mathcal{N}(A\theta, I_2)$; prior $\theta \sim \mathcal{N}(0, I_2)$. Minimize $J = \frac{1}{2}\|b - A\theta\|^2 + \frac{1}{2}\|\theta\|^2$ in these numerical coordinates; I_2 is the identity.

A. The prior makes the two data columns independent.

B. Gaussian noise makes every data fit unique.

C. The prior adds curvature along the data null space.

MAP estimate; positive definiteness — [1] DL §5.6.1 · [1] DL §2.7 · [1] DL §4.3 · Course question

SOLUTION

Estimation

What makes the MAP estimate unique?

A null-space direction leaves predictions unchanged.

A: fixed 2×2 design; b: observed 2-entry column; θ: parameter 2-entry column; $r = b - A\theta$: reading residual; $I = I_2$: identity; J: objective. θ_1, θ_2: the two coefficients.

---

1. $\theta = (\theta_1, \theta_2)^T$: parameters; $b = (1, 2)^T$: readings.
2. $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$; $r = b - A\theta$: residual.
3. $A\theta = (\theta_1 + \theta_2)(1, 2)^T$; zero residual requires $\theta_1 + \theta_2 = 1$.

MAP estimate; positive definiteness — [1] DL §5.6.1 · [1] DL §2.7 · [1] DL §4.3 · Course question

SOLUTION

Estimation

What makes the MAP estimate unique?

Add the stated prior penalty

A: fixed 2×2 design; b: observed 2-entry column; θ: parameter 2-entry column; $r = b - A\theta$: reading residual; $I = I_2$: identity; J: objective. λ^2: prior weight; σ^2, τ^2: noise/prior variances, with ratio 1.

1. J: scaled negative-log posterior; $J = \frac{1}{2}\|r\|^2 + \frac{1}{2}\|\theta\|^2$, with $\lambda^2 = \sigma^2/\tau^2 = 1$.
2. $\nabla J = A^T(A\theta - b) + \theta = 0$.
3. $A^T A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$, $A^T b = (5, 5)^T$; hence $(A^T A + I)\theta = A^T b$.

MAP estimate; positive definiteness — [1] DL §5.6.1 · [1] DL §2.7 · [1] DL §4.3 · Course question

SOLUTION

Estimation

What makes the MAP estimate unique?

Solution 3 / 5

Solve the two equations

A: fixed 2×2 design; b: observed 2-entry column; θ: parameter 2-entry column; $r = b - A\theta$: reading residual; $I = I_2$: identity; J: objective. $\hat{\theta}_{MAP}$: fitted posterior-mode column.

---

1. $6\theta_1 + 5\theta_2 = 5$; $5\theta_1 + 6\theta_2 = 5$.
2. Subtract: $\theta_1 - \theta_2 = 0$; substitute: $11\theta_1 = 5$.
3. $\hat{\theta}_{MAP} = (5/11, 5/11)^T$; $r = (1/11, 2/11)^T$.

MAP estimate; positive definiteness — [1] DL §5.6.1 · [1] DL §2.7 · [1] DL §4.3 · Course question

SOLUTION

Estimation

What makes the MAP estimate unique?

Solution 4 / 5

The Hessian collects the objective’s second partial derivatives.

A: fixed 2×2 design; b: observed 2-entry column; θ: parameter 2-entry column; $r = b - A\theta$: reading residual; $I = I_2$: identity; J: objective. $v \in \mathbb{R}^2 \setminus \{0\}$: parameter direction; Hessian: second-derivative matrix.

1. For any nonzero column v, $v^T(A^TA + I)v = \|Av\|^2 + \|v\|^2 > 0$.
2. For the data-null direction $v = (1, -1)^T$, $Av = 0$ but $\|v\|^2 = 2$.
3. Thus the total objective has positive curvature even where the data loss is flat.

MAP estimate; positive definiteness — [1] DL §5.6.1 · [1] DL §2.7 · [1] DL §4.3 · Course question

SOLUTION

Estimation

What makes the MAP estimate unique?

Solution 5 / 5

Choose C

A: fixed 2×2 design; b: observed 2-entry column; θ: parameter 2-entry column; $r = b - A\theta$: reading residual; $I = I_2$: identity; J: objective.

1. A is false: A still has identical columns and rank 1.
2. B is false: Gaussian noise alone does not identify their separate coefficients.
3. C is correct: the stated Gaussian prior supplies curvature in the null direction.

MAP estimate; positive definiteness — [1] DL §5.6.1 · [1] DL §2.7 · [1] DL §4.3 · Course question

RESULT

Estimation

What makes the MAP estimate unique?

$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ known; $b = (1,2)^T$ observed; $\theta \in \mathbb{R}^2$. Model: $b \mid \theta \sim \mathcal{N}(A\theta, I_2)$; prior $\theta \sim \mathcal{N}(0, I_2)$. Minimize $J = \frac{1}{2}\|b - A\theta\|^2 + \frac{1}{2}\|\theta\|^2$ in these numerical coordinates; I_2 is the identity.

A The prior makes the two data columns independent.

B Gaussian noise makes every data fit unique.

C The prior adds curvature along the data null space. ✓

> Least squares: $\theta_1 + \theta_2 = 1$. MAP: $(A^TA + I)\theta = A^Tb$ gives $\theta_1 = \theta_2 = 5/11$.

MAP estimate; positive definiteness — [1] DL §5.6.1 · [1] DL §2.7 · [1] DL §4.3 · Course question

EXERCISE

Sensor Calibration

Calibration: choose the likelihood

$b = (1, 2, 3)^T$: readings minus a known reference; $b_i = \theta + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Which data objective fits this model? $i = 1, 2, 3$: reading index; b: observed 3-entry column; θ: scalar offset; ε_i: random error in reading units.

A $\frac{1}{2} \sum_i (b_i - \theta)^2$

B $\sum_i |b_i - \theta|$

C $\frac{1}{2} \sum_i (b_i - \theta)^2 + \frac{1}{2}\theta^2$

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

SOLUTION

Sensor Calibration

Calibration: choose the likelihood

Solution 1 / 4

Define the calibration model

$b = (b_1, b_2, b_3)^T$: fixed data; i: reading index; θ: scalar offset; ε_i: zero-mean random noise. J_{data}: scalar data loss; C: constant in θ; $\hat{\theta}_{MLE}$: data-only estimate; $\bar{b}$: data average.

---

1. $b = (1, 2, 3)^T$: observed readings; θ: unknown common offset.
2. $b_i = \theta + \varepsilon_i$; independent $\varepsilon_i \sim \mathcal{N}(0, 1)$.
3. $p(b_i \mid \theta) = (2\pi)^{-1/2} \exp[-(b_i - \theta)^2/2]$.

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

SOLUTION

Sensor Calibration

Calibration: choose the likelihood

From likelihood to objective

$b = (b_1, b_2, b_3)^T$: fixed data; i: reading index; θ: scalar offset; ε_i: zero-mean random noise. J_{data}: scalar data loss; C: constant in θ; $\hat{\theta}_{MLE}$: data-only estimate; $\bar{b}$: data average.

---

1. Independence: $p(b \mid \theta) = \prod_{i=1}^{3} p(b_i \mid \theta)$.
2. $-\log p(b \mid \theta) = C + \frac{1}{2} \sum_{i=1}^{3} (b_i - \theta)^2$.
3. C is independent of θ; minimizing leaves $J_{data} = \frac{1}{2} \sum_i (b_i - \theta)^2$.

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

SOLUTION

Sensor Calibration

Calibration: choose the likelihood

Solution 3 / 4

Expand every square

$b = (b_1, b_2, b_3)^T$: fixed data; i: reading index; θ: scalar offset; ε_i: zero-mean random noise. J_{data}: scalar data loss; C: constant in θ; $\hat{\theta}_{MLE}$: data-only estimate; $\bar{b}$: data average.

---

1. $2J_{data} = (1-\theta)^2 + (2-\theta)^2 + (3-\theta)^2.$
2. $= (1 - 2\theta + \theta^2) + (4 - 4\theta + \theta^2) + (9 - 6\theta + \theta^2).$
3. $= 3\theta^2 - 12\theta + 14.$

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

SOLUTION

Sensor Calibration

Calibration: choose the likelihood

Solution 4 / 4

Minimize and choose A

$b = (b_1, b_2, b_3)^T$: fixed data; i: reading index; θ: scalar offset; ε_i: zero-mean random noise. J_{data}: scalar data loss; C: constant in θ; $\hat{\theta}_{MLE}$: data-only estimate; $\bar{b}$: data average.

---

1. $J'_{data} = 3\theta - 6 = 0 \Rightarrow \hat{\theta}_{MLE} = 2;\ J''_{data} = 3 > 0.$
2. A follows from independent Gaussian noise.
3. B uses absolute residuals, a different noise model. C already adds a prior penalty.

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

RESULT

Sensor Calibration

Calibration: choose the likelihood

$b = (1, 2, 3)^T$: readings minus a known reference; $b_i = \theta + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Which data objective fits this model? $i = 1, 2, 3$: reading index; b: observed 3-entry column; θ: scalar offset; ε_i: random error in reading units.

A $\frac{1}{2}\sum_i (b_i - \theta)^2$ ✓

B $\sum_i |b_i - \theta|$

C $\frac{1}{2}\sum_i (b_i - \theta)^2 + \frac{1}{2}\theta^2$

$-\log p(b \mid \theta) = C + \frac{1}{2}\sum_i (b_i - \theta)^2$; $\hat{\theta}_{MLE} = \bar{b} = 2$.

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

EXERCISE

Sensor Calibration

Calibration: combine data and prior

$b = (1, 2, 3)^T$ observed reference-subtracted readings; independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$, $i = 1, 2, 3$; prior $\theta \sim \mathcal{N}(0, 1)$. θ: scalar offset; normal notation gives mean, variance; variance 1 is known, in squared reading units. Find posterior mean and precision (inverse variance).

A. Center stays at 2; only uncertainty changes.

B. Center moves toward 0; precision becomes 4.

C. Center moves toward 0; variances add to 4.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

SOLUTION

Sensor Calibration

Calibration: combine data and prior

Keep the data and state the prior

b: fixed 3-entry column; θ: scalar offset; J_{data}, J_{post}: data/negative-log posterior objectives. μ_1: posterior mean; P_1: posterior variance (squared reading units); $1/P_1$: precision; $\theta \mid b$: offset distribution given data.

1. $b = (1, 2, 3)^T$; independent noise variance 1; θ: offset.
2. $J_{data} = \frac{1}{2}(3\theta^2 - 12\theta + 14)$; $\theta \sim \mathcal{N}(0, 1)$.
3. Negative log prior adds $\theta^2/2$, up to a constant.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

SOLUTION

Sensor Calibration

Calibration: combine data and prior

Collect the posterior quadratic

b: fixed 3-entry column; θ: scalar offset; J_{data}, J_{post}: data/negative-log posterior objectives. μ_1: posterior mean; P_1: posterior variance (squared reading units); $1/P_1$: precision; $\theta \mid b$: offset distribution given data.

1. $J_{post} = \frac{1}{2}(3\theta^2 - 12\theta + 14) + \frac{1}{2}\theta^2.$
2. $= 2\theta^2 - 6\theta + 7 = 2(\theta^2 - 3\theta) + 7.$
3. $\theta^2 - 3\theta = (\theta - 3/2)^2 - 9/4.$

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

SOLUTION

Sensor Calibration

Calibration: combine data and prior

Complete the square and read variance

b: fixed 3-entry column; θ: scalar offset; J_{data}, J_{post}: data/negative-log posterior objectives. μ_1: posterior mean; P_1: posterior variance (squared reading units); $1/P_1$: precision; $\theta \mid b$: offset distribution given data.

1. $J_{post} = 2(\theta - 3/2)^2 - 9/2 + 7 = 2(\theta - 1.5)^2 + 2.5.$
2. Gaussian quadratic: $(\theta - \mu_1)^2/(2P_1)$; hence $\mu_1 = 1.5$, $1/(2P_1) = 2$.
3. $P_1 = 1/4$; precision $P_1^{-1} = 4$; $\theta \mid b \sim \mathcal{N}(1.5, 1/4)$.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

SOLUTION

Sensor Calibration

Calibration: combine data and prior

Solution 4 / 4

Choose B and check the weights

b: fixed 3-entry column; θ: scalar offset; J_{data}, J_{post}: data/negative-log posterior objectives. μ_1: posterior mean; P_1: posterior variance (squared reading units); $1/P_1$: precision; $\theta \mid b$: offset distribution given data.

1. Data precision 3 plus prior precision 1 gives 4.
2. Posterior mean $= (3 \cdot 2 + 1 \cdot 0)/4 = 1.5$; MAP equals this Gaussian mean.
3. B is correct. A ignores the prior shift. C adds variances instead of precisions.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

RESULT

Sensor Calibration

Calibration: combine data and prior

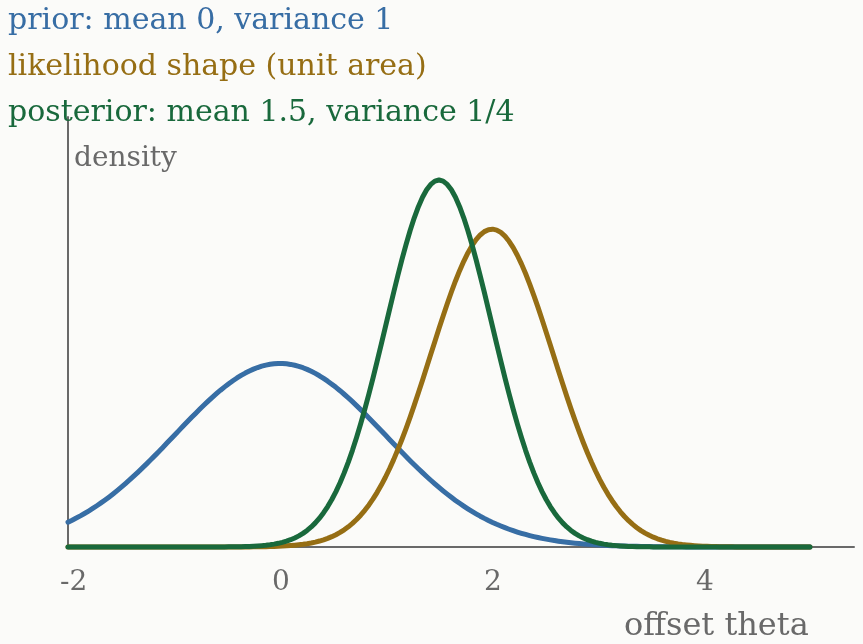
$b = (1, 2, 3)^T$ observed reference-subtracted readings; independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$, $i = 1, 2, 3$; prior $\theta \sim \mathcal{N}(0, 1)$. θ : scalar offset; normal notation gives mean, variance; variance 1 is known, in squared reading units. Find posterior mean and precision (inverse variance).

A Center stays at 2; only uncertainty changes.

B Center moves toward 0; precision becomes 4. ✓

C Center moves toward 0; variances add to 4.

$J_{post} = 2(\theta - 1.5)^2 + 2.5$; $\theta \mid b \sim \mathcal{N}(1.5, 1/4)$.



Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

EXERCISE

Sensor Calibration

Calibration: confidence coverage

Three independent $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$; mean θ, known variance 1 in squared reading units. θ: fixed scalar offset; $B = (B_1, B_2, B_3)^T$: random data; $\bar{B} = (B_1 + B_2 + B_3)/3$. $C(B) = \bar{B} \pm 1.96/\sqrt{3}$, in reading units. Interpret 95% confidence.

A. This realized interval has 95% posterior mass.

B. 95% of future individual readings lie inside it.

C. Repeated intervals cover fixed θ 95% of the time.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

SOLUTION

Sensor Calibration

Calibration: confidence coverage

Define the random interval

$i = 1, 2, 3$; B_i: random reading; B: random 3-entry column; $\bar{B}$: average; θ: fixed offset. $C(B)$: random confidence interval; Z: dimensionless standardized mean; P_θ: repeated-data probability with θ held fixed.

1. θ: fixed true offset; B_i: random reading before observation.
2. $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$ independently; $\bar{B} = (B_1 + B_2 + B_3)/3$.
3. $C(B)$: confidence interval computed anew for each random dataset.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

SOLUTION

Sensor Calibration

Calibration: confidence coverage

Standard error is the sample mean’s sampling standard deviation.

$i = 1, 2, 3$; B_i: random reading; B: random 3-entry column; $\bar{B}$: average; θ: fixed offset. $C(B)$: random confidence interval; Z: dimensionless standardized mean; P_θ: repeated-data probability with θ held fixed.

1. $\mathbb{E}_\theta[\bar{B}] = \theta$; $\mathrm{Var}_\theta(\bar{B}) = (1+1+1)/9 = 1/3$.
2. $Z = (\bar{B} - \theta)/(1/\sqrt{3}) \sim \mathcal{N}(0, 1)$.
3. $P_\theta(-1.96 \le Z \le 1.96) \approx .95$.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

SOLUTION

Sensor Calibration

Calibration: confidence coverage

Solution 3 / 4

A confidence interval comes from a rule with the stated repeated-sampling coverage.

$i = 1, 2, 3$; B_i: random reading; B: random 3-entry column; $\bar{B}$: average; θ: fixed offset. $C(B)$: random confidence interval; Z: dimensionless standardized mean; P_θ: repeated-data probability with θ held fixed.

1. $-1.96/\sqrt{3} \leq \bar{B} - \theta \leq 1.96/\sqrt{3}$.
2. $\bar{B} - 1.96/\sqrt{3} \leq \theta \leq \bar{B} + 1.96/\sqrt{3}$.
3. Thus $C(B) = \bar{B} \pm 1.96/\sqrt{3}$ and $P_\theta(\theta \in C(B)) \approx .95$.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

SOLUTION

Sensor Calibration

Calibration: confidence coverage

Solution 4 / 4

Choose C

$i = 1, 2, 3$; B_i: random reading; B: random 3-entry column; $\bar{B}$: average; θ: fixed offset. $C(B)$: random confidence interval; Z: dimensionless standardized mean; P_θ: repeated-data probability with θ held fixed.

1. C: repeated random intervals cover the same fixed offset about 95% of the time.
2. A needs a posterior distribution; confidence alone provides none.
3. B concerns individual future readings, whose noise variance is 1, not 1/3.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

RESULT

Sensor Calibration

Calibration: confidence coverage

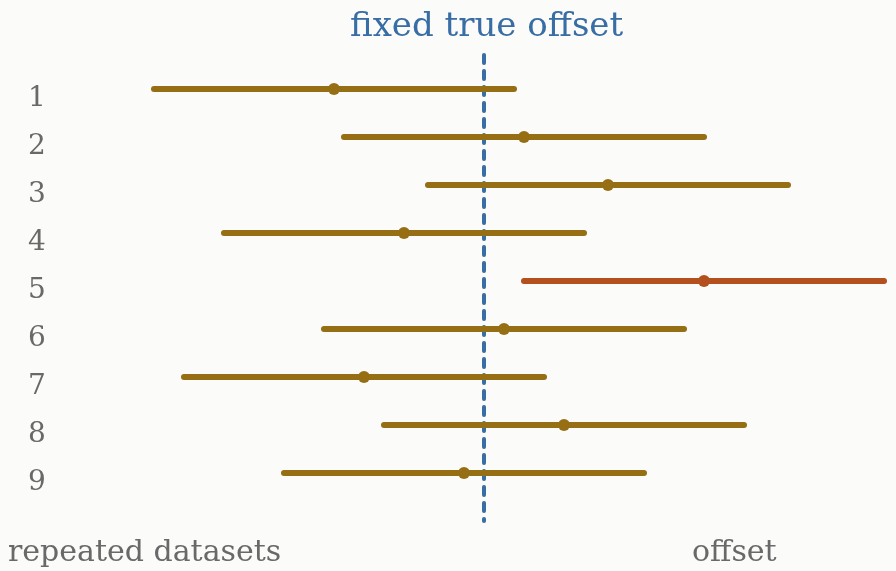
Three independent $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$; mean θ, known variance 1 in squared reading units. θ: fixed scalar offset; $B = (B_1, B_2, B_3)^T$: random data; $\bar{B} = (B_1 + B_2 + B_3)/3$. $C(B) = \bar{B} \pm 1.96/\sqrt{3}$, in reading units. Interpret 95% confidence.

A. This realized interval has 95% posterior mass.

B. 95% of future individual readings lie inside it.

C. Repeated intervals cover fixed θ 95% of the time. ✓

$\sqrt{3}(\bar{B} - \theta) \sim \mathcal{N}(0, 1)$; $P_\theta(\theta \in C(B)) \approx .95$.



Illustration, not measured coverage

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

EXERCISE

Sensor Calibration

Calibration: what does credibility describe?

Observed $b = (1, 2, 3)^T$; independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$; prior $\theta \sim \mathcal{N}(0, 1)$. Hence $\theta \mid b \sim \mathcal{N}(1.5, .25)$. θ: scalar offset; b: fixed 3-entry column; normal notation gives mean, variance; offset uses reading units and variance squared units. What does a 95% credible interval describe?

A. 95% posterior mass for θ, with b fixed.

B. 95% coverage for every fixed true offset.

C. 95% probability for a future noisy reading.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

SOLUTION

Sensor Calibration

Calibration: what does credibility describe?

Fix the data and use the posterior

b: fixed 3-entry column; θ: scalar offset; μ_1, P_1: posterior mean/variance; $I(b)$: interval from fixed data; Z: dimensionless standardized offset. $P(\cdot \mid b)$: probability under the stated posterior.

1. $b = (1, 2, 3)^T$: observed data, now fixed; θ: offset.
2. $\theta \mid b \sim \mathcal{N}(\mu_1, P_1)$ with $\mu_1 = 1.5$, $P_1 = .25$.
3. Posterior standard deviation $\sqrt{P_1} = \sqrt{.25} = .5$.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

SOLUTION

Sensor Calibration

Calibration: what does credibility describe?

Standardize, then undo the scaling

b: fixed 3-entry column; θ: scalar offset; μ_1, P_1: posterior mean/variance; $I(b)$: interval from fixed data; Z: dimensionless standardized offset. $P(\cdot \mid b)$: probability under the stated posterior.

1. $Z = (\theta - 1.5)/.5 \mid b \sim \mathcal{N}(0, 1).$
2. $P(-1.96 \leq Z \leq 1.96 \mid b) \approx .95.$
3. $1.5 - 1.96(.5) \leq \theta \leq 1.5 + 1.96(.5)$ gives $I(b) = [.52, 2.48].$

Posterior credible interval — [4] MIT 18.05 §2 · Course example

SOLUTION

Sensor Calibration

Calibration: what does credibility describe?

Solution 3 / 3

A credible interval contains the stated posterior probability.

b: fixed 3-entry column; θ: scalar offset; μ_1, P_1: posterior mean/variance; $I(b)$: interval from fixed data; Z: dimensionless standardized offset. $P(\cdot \mid b)$: probability under the stated posterior.

---

1. A: $P(\theta \in I(b) \mid b) \approx .95$ under this posterior.
2. B is a repeated-sampling claim for every fixed offset; it does not follow here.
3. C includes new measurement noise; this interval describes the offset only.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

RESULT

Sensor Calibration

Calibration: what does credibility describe?

Observed $b = (1, 2, 3)^T$; independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$; prior $\theta \sim \mathcal{N}(0, 1)$. Hence $\theta \mid b \sim \mathcal{N}(1.5, .25)$. θ: scalar offset; b: fixed 3-entry column; normal notation gives mean, variance; offset uses reading units and variance squared units. What does a 95% credible interval describe?

- **A** 95% posterior mass for θ, with b fixed. ✓
- **B** 95% coverage for every fixed true offset.
- **C** 95% probability for a future noisy reading.

> $I(b) = 1.5 \pm 1.96(.5)$; $P(\theta \in I(b) \mid b) \approx .95$.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

EXERCISE

Sensor Calibration

Calibration: why do the intervals differ?

$b = (1, 2, 3)^T$ observed; independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$; prior $\theta \sim \mathcal{N}(0, 1)$. θ: scalar offset; both stated variances are known 1 in squared reading units. Confidence $C(b) = 2 \pm 1.96/\sqrt{3}$; credible $I(b) = 1.5 \pm 1.96(.5)$, in reading units. Why different?

A. Narrower credible interval means better coverage.

B. The zero-centered prior adds information.

C. The physical sensor became less noisy.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

SOLUTION

Sensor Calibration

Calibration: why do the intervals differ?

Identify the two information sources

b: observed 3-entry column; $\bar{b}$: data average; θ: scalar offset; $C(b)$: realized confidence interval; $I(b)$: credible interval. Precision is inverse variance; endpoints use reading units, variances squared units.

1. $b = (1, 2, 3)^T$; $\bar{b} = 2$; noise variance 1; prior $\mathcal{N}(0, 1)$.
2. C: confidence interval; sampling standard error $1/\sqrt{3}$.
3. I: credible interval; posterior mean $(3 \cdot 2 + 1 \cdot 0)/4 = 1.5$, variance $1/4$.
4. Data precision $= 1/(1/3) = 3$; prior precision $= 1/1 = 1$.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

SOLUTION

Sensor Calibration

Calibration: why do the intervals differ?

Solution 2 / 3

Compute and compare endpoints

b: observed 3-entry column; $\bar{b}$: data average; θ: scalar offset; $C(b)$: realized confidence interval; $I(b)$: credible interval. Precision is inverse variance; endpoints use reading units, variances squared units.

---

1. $C = 2 \pm 1.96/\sqrt{3} \approx 2 \pm 1.1316 = [.8684, 3.1316]$.
2. $I = 1.5 \pm 1.96\sqrt{1/4} = 1.5 \pm .98 = [.52, 2.48]$.
3. $1.5 < 2$ and $.98 < 1.1316$: the prior shifts the center and reduces posterior width.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

SOLUTION

Sensor Calibration

Calibration: why do the intervals differ?

Solution 3 / 3

Choose B

b: observed 3-entry column; $\bar{b}$: data average; θ: scalar offset; $C(b)$: realized confidence interval; $I(b)$: credible interval. Precision is inverse variance; endpoints use reading units, variances squared units.

1. B: one unit of prior precision adds to three units from the data.
2. A: narrower does not prove better repeated-sampling coverage.
3. C: the physical noise variance remains 1; only the information used changed.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

RESULT

Sensor Calibration

Calibration: why do the intervals differ?

$b = (1, 2, 3)^T$ observed; independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$; prior $\theta \sim \mathcal{N}(0, 1)$. θ: scalar offset; both stated variances are known 1 in squared reading units. Confidence $C(b) = 2 \pm 1.96/\sqrt{3}$; credible $I(b) = 1.5 \pm 1.96(.5)$, in reading units. Why different?

A Narrower credible interval means better coverage.

B The zero-centered prior adds information. ✓

C The physical sensor became less noisy.

$C \approx [.8684, 3.1316]$; $I \approx [.5200, 2.4800]$. Prior precision 1 adds to data precision 3.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

EXERCISE

Sensor Calibration

Calibration: new data, new prior

$b = (2, 3)^T$: observed reference-subtracted readings. Independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$, $i = 1, 2$; prior $\theta \sim \mathcal{N}(4, 1)$. θ: scalar offset in reading units; normal notation gives mean, variance; variance 1 is known in squared units. Which is valid?

A. The prior changes the data-only MLE.

B. The prior center can be ignored in MAP.

C. MAP moves above the data mean; MLE stays fixed.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

SOLUTION

Sensor Calibration

Calibration: new data, new prior

Set up the new data-only problem

b: fixed 2-entry column; θ: scalar offset; J_{data}, J_{post}: data/posterior objectives; $\hat{\theta}_{MLE}, \hat{\theta}_{MAP}$: fitted offsets. μ_1, P_1: posterior mean/variance; C, I: confidence/credible intervals; endpoints use reading units.

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1. $b = (2,3)^T$; θ: offset; independent noise variance 1.
2. $J_{data} = \frac{1}{2}[(2-\theta)^2 + (3-\theta)^2]$.
3. $= \frac{1}{2}[(4 - 4\theta + \theta^2) + (9 - 6\theta + \theta^2)] = \frac{1}{2}(2\theta^2 - 10\theta + 13)$.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

SOLUTION

Sensor Calibration

Calibration: new data, new prior

Solution 2 / 5

Minimize, then add the new prior

b: fixed 2-entry column; θ: scalar offset; J_{data}, J_{post}: data/posterior objectives; $\hat{\theta}_{MLE}, \hat{\theta}_{MAP}$: fitted offsets. μ_1, P_1: posterior mean/variance; C, I: confidence/credible intervals; endpoints use reading units.

1. $J'_{data} = 2\theta - 5 = 0 \Rightarrow \hat{\theta}_{MLE} = 2.5;\ J''_{data} = 2 > 0.$
2. Prior $\mathcal{N}(4, 1)$ adds $\frac{1}{2}(\theta - 4)^2 = \frac{1}{2}(\theta^2 - 8\theta + 16).$
3. $J_{post} = \frac{1}{2}(3\theta^2 - 18\theta + 29).$

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

SOLUTION

Sensor Calibration

Calibration: new data, new prior

Complete the posterior square

b: fixed 2-entry column; θ: scalar offset; J_{data}, J_{post}: data/posterior objectives; $\hat{\theta}_{MLE}, \hat{\theta}_{MAP}$: fitted offsets. μ_1, P_1: posterior mean/variance; C, I: confidence/credible intervals; endpoints use reading units.

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1. $3\theta^2 - 18\theta + 29 = 3(\theta^2 - 6\theta) + 29.$
2. $= 3[(\theta - 3)^2 - 9] + 29 = 3(\theta - 3)^2 + 2.$
3. $J_{post} = \frac{3}{2}(\theta - 3)^2 + 1;\ \mu_1 = 3,\ 1/(2P_1) = 3/2 \Rightarrow P_1 = 1/3.$

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

SOLUTION

Sensor Calibration

Calibration: new data, new prior

Compute both interval constructions

b: fixed 2-entry column; θ: scalar offset; J_{data}, J_{post}: data/posterior objectives; $\hat{\theta}_{MLE}, \hat{\theta}_{MAP}$: fitted offsets. μ_1, P_1: posterior mean/variance; C, I: confidence/credible intervals; endpoints use reading units.

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1. C: confidence; mean 2.5, sampling standard error $1/\sqrt{2}$.
2. $C = 2.5 \pm 1.96/\sqrt{2} \approx [1.1141, 3.8859]$.
3. I: credible; posterior mean 3, standard deviation $1/\sqrt{3}$; $I = 3 \pm 1.96/\sqrt{3} \approx [1.8684, 4.1316]$.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

SOLUTION

Sensor Calibration

Calibration: new data, new prior

Solution 5 / 5

Choose C

b: fixed 2-entry column; θ: scalar offset; J_{data}, J_{post}: data/posterior objectives; $\hat{\theta}_{MLE}, \hat{\theta}_{MAP}$: fitted offsets. μ_1, P_1: posterior mean/variance; C, I: confidence/credible intervals; endpoints use reading units.

1. C: $\hat{\theta}_{MAP} = 3 > 2.5$, while the data-only MLE remains 2.5.
2. A incorrectly puts prior information into the data-only MLE.
3. B drops the prior center 4 and its linear term. Confidence and credibility retain different meanings.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

RESULT

Sensor Calibration

Calibration: new data, new prior

$b = (2, 3)^T$: observed reference-subtracted readings. Independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$, $i = 1, 2$; prior $\theta \sim \mathcal{N}(4, 1)$. θ: scalar offset in reading units; normal notation gives mean, variance; variance 1 is known in squared units. Which is valid?

A The prior changes the data-only MLE.

B The prior center can be ignored in MAP.

C MAP moves above the data mean; MLE stays fixed. ✓

> $\hat{\theta}_{MLE} = 2.5$, $\theta \mid b \sim \mathcal{N}(3, 1/3)$. $C = 2.5 \pm 1.96/\sqrt{2}$; $I = 3 \pm 1.96/\sqrt{3}$.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

EXERCISE

Sensor Calibration

What does the sensor promise?

A specification says “±2 cm, 95%.” Which conclusion is justified without another definition?

A. It is a confidence interval for the offset.

B. It is a credible interval for the offset.

C. None: ask what quantity and probability model it describes.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

SOLUTION

Sensor Calibration

What does the sensor promise?

Identify what the specification omits

θ: scalar offset; B: random data column; b: observed data; $C(B)$: confidence interval; $I(b)$: credible interval. ε_{new}: fresh scalar noise; $Y = \theta + \varepsilon_{new}$: next reference-subtracted reading; lengths use cm.

1. The statement "± 2 cm, 95%" supplies a half-width of 2 cm and a percentage.
2. θ: offset; B: repeated random dataset; b: observed dataset.
3. No target quantity or probability construction is specified.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

SOLUTION

Sensor Calibration

What does the sensor promise?

Compare three possible meanings

θ: scalar offset; B: random data column; b: observed data; $C(B)$: confidence interval; $I(b)$: credible interval. ε_{new}: fresh scalar noise; $Y = \theta + \varepsilon_{new}$: next reference-subtracted reading; lengths use cm.

1. Confidence: $P_\theta(\theta \in C(B)) \approx .95$ for a stated repeated-data procedure.
2. Credibility: $P(\theta \in I(b) \mid b) \approx .95$ under a stated posterior.
3. Fresh-noise bound: $P(|\varepsilon_{new}| \leq 2\ \text{cm}) \approx .95$ under a stated noise model.
4. For $Y = \theta + \varepsilon_{new}$ with finite variances and ε_{new} independent of (θ, b): $\text{Var}(Y \mid b) = \text{Var}(\theta \mid b) + \text{Var}(\varepsilon_{new})$.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

SOLUTION

Sensor Calibration

What does the sensor promise?

Solution 3 / 3

Choose C

θ: scalar offset; B: random data column; b: observed data; $C(B)$: confidence interval; $I(b)$: credible interval. ε_{new}: fresh scalar noise; $Y = \theta + \varepsilon_{new}$: next reference-subtracted reading; lengths use cm.

1. A assumes an unstated repeated-sampling procedure.
2. B assumes an unstated prior and posterior.
3. C: ask which quantity is bounded and which model or procedure supplies 95%.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

RESULT

Sensor Calibration

What does the sensor promise?

A specification says “±2 cm, 95%.” Which conclusion is justified without another definition?

A. It is a confidence interval for the offset.

B. It is a credible interval for the offset.

C. None: ask what quantity and probability model it describes. ✓

> Offset uncertainty, future-reading error and repeated coverage are different claims.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

EXERCISE

Synthesis

Which modeling bridge is valid?

$b \mid \theta \sim \mathcal{N}(A\theta, \sigma^2 I_N)$; prior $\theta \sim \mathcal{N}(\mu_0, \tau^2 I_p)$. A: known N×p design; b: observed N-column; θ, μ_0: parameter/fixed prior-mean p-columns; known $\sigma^2, \tau^2 > 0$. Errors are independent, zero-mean Gaussians; variances use their respective squared units. Which chain gives the negative-log posterior?

A. Independence gives squares; Gaussian noise gives sums.

B. Gaussian noise gives squares; $-\log$ turns independent factors into sums.

C. A Gaussian prior removes the data residual term.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

SOLUTION

Synthesis

Which modeling bridge is valid?

Solution 1 / 4

State the model and symbols

N, p: reading/parameter counts; $A \in \mathbb{R}^{N \times p}$: fixed design; $b, r, \varepsilon \in \mathbb{R}^N$: data/residual/noise; $\theta, \mu_0 \in \mathbb{R}^p$: parameter/prior mean; I_p: identity. i: index; C: constant in θ; J: objective. $\sigma^2, \tau^2 > 0$: noise/prior variances.

1. $b = A\theta + \varepsilon$: data column b, design matrix A, parameter column θ.
2. ε_i are independent $\mathcal{N}(0, \sigma^2)$ noises; $r = b - A\theta$: residual.
3. Prior $\theta \sim \mathcal{N}(\mu_0, \tau^2 I_p)$; μ_0: prior mean; $\sigma^2, \tau^2 > 0$: variances.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

SOLUTION

Synthesis

Which modeling bridge is valid?

Solution 2 / 4

Separate the square from the sum

N, p: reading/parameter counts; $A \in \mathbb{R}^{N \times p}$: fixed design; $b, r, \varepsilon \in \mathbb{R}^N$: data/residual/noise; $\theta, \mu_0 \in \mathbb{R}^p$: parameter/prior mean; I_p: identity. i: index; C: constant in θ; J: objective. r_i: residual entry; $p(\cdot \mid \theta)$: data density.

1. One Gaussian factor: $p(b_i \mid \theta) = (2\pi\sigma^2)^{-1/2} e^{-r_i^2/(2\sigma^2)}$.
2. Conditional independence given θ: $p(b \mid \theta) = \prod_i p(b_i \mid \theta)$.
3. $-\log p(b \mid \theta) = C + \sum_i r_i^2/(2\sigma^2) = C + \|b - A\theta\|^2/(2\sigma^2)$.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

SOLUTION

Synthesis

Which modeling bridge is valid?

Solution 3 / 4

Add the prior through Bayes rule

N, p: reading/parameter counts; $A \in \mathbb{R}^{N \times p}$: fixed design; $b, r, \varepsilon \in \mathbb{R}^N$: data/residual/noise; $\theta, \mu_0 \in \mathbb{R}^p$: parameter/prior mean; I_p: identity. i: index; C: constant in θ; J: objective. $\lambda^2 = \sigma^2/\tau^2$: prior weight; $\propto$: equality up to a factor independent of θ.

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1. $p(\theta \mid b) \propto p(b \mid \theta)p(\theta)$; logarithms turn this product into a sum.
2. $J = \|b - A\theta\|^2/(2\sigma^2) + \|\theta - \mu_0\|^2/(2\tau^2)$.
3. Multiply all of J by σ^2: $\frac{1}{2}\|r\|^2 + \frac{\lambda^2}{2}\|\theta - \mu_0\|^2$, where $\lambda^2 = \sigma^2/\tau^2$.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

SOLUTION

Synthesis

Which modeling bridge is valid?

Solution 4 / 4

Choose B

N, p: reading/parameter counts; $A \in \mathbb{R}^{N \times p}$: fixed design; $b, r, \varepsilon \in \mathbb{R}^N$: data/residual/noise; $\theta, \mu_0 \in \mathbb{R}^p$: parameter/prior mean; I_p: identity. i: index; C: constant in θ; J: objective.

1. B: the Gaussian exponent supplies squares; independence and logarithms supply the sum.
2. A reverses the two reasons.
3. C discards the likelihood. A prior adds a penalty while retaining the residual term.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

RESULT

Synthesis

Which modeling bridge is valid?

$b \mid \theta \sim \mathcal{N}(A\theta, \sigma^2 I_N)$; prior $\theta \sim \mathcal{N}(\mu_0, \tau^2 I_p)$. A: known N×p design; b: observed N-column; θ, μ_0: parameter/fixed prior-mean p-columns; known $\sigma^2, \tau^2 > 0$. Errors are independent, zero-mean Gaussians; variances use their respective squared units. Which chain gives the negative-log posterior?

A. Independence gives squares; Gaussian noise gives sums.

B. Gaussian noise gives squares; $-\log$ turns independent factors into sums. ✓

C. A Gaussian prior removes the data residual term.

> $$-\log p(\theta \mid b) = C + \|b - A\theta\|^2/(2\sigma^2) + \|\theta - \mu_0\|^2/(2\tau^2).$$

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question